

Example:

$G=(V_N, V_T, P, S)$ $V_N=\{S, A, B, C, D\}$ $V_T=\{a, b\}$

$P=\{1. S \rightarrow AC$

2. $S \rightarrow bA$

3. $S \rightarrow B$

4. $S \rightarrow aA$

5. $A \rightarrow \epsilon$

6. $A \rightarrow aS$

7. $A \rightarrow ABAb$

8. $B \rightarrow a$

9. $B \rightarrow AbSA$

10. $C \rightarrow abC$

11. $D \rightarrow AB\}$

1. Elimination of ϵ productions:

a) $N_\epsilon = \emptyset$

b) for the production $A \rightarrow \epsilon$ $N_\epsilon = \emptyset \cup \{A\}$

$$N_\epsilon = \{A\}$$

$P'=\{1. S \rightarrow AC$

2. $S \rightarrow bA$

3. $S \rightarrow B$

4. $S \rightarrow aA$

5. $A \rightarrow aS$

6. $A \rightarrow ABAb$

7. $B \rightarrow a$

8. $B \rightarrow AbSA$

9. $C \rightarrow abC$

10. $D \rightarrow AB$

11. $S \rightarrow C$

12. $S \rightarrow b$

13. $S \rightarrow a$

14. $A \rightarrow BAb$

15. $A \rightarrow ABb$

16. $A \rightarrow Bb$

17. $B \rightarrow bSA$

18. $B \rightarrow AbS$

19. $B \rightarrow bS$

20. $D \rightarrow B \}$

2. Elimination of renaming- Unit Productions:

The production that has the form $X \rightarrow Y$, X and Y are nonterminal, is called renaming.

The renaming from P' are: $S \rightarrow B$, $S \rightarrow C$, $D \rightarrow B$

$$R_S = \{S\}, R_B = \{B\}, R_C = \{C\}, R_D = \{D\}$$

$$\text{for } S \rightarrow B \quad R_B = R_B \cup R_S = \{B\} \cup \{S\} = \{B, S\}$$

$$\text{for } S \rightarrow C \quad R_C = R_C \cup R_S = \{C\} \cup \{S\} = \{C, S\}$$

$$\text{for } D \rightarrow B \quad R_B = R_B \cup R_D = \{B, S\} \cup \{D\} = \{B, S, D\}$$

$P''=\{1. S \rightarrow AC$

2. $S \rightarrow bA$

3. $S \rightarrow aA$

4. $A \rightarrow aS$

5. $A \rightarrow ABAb$

6. $B \rightarrow a$

18. $S \rightarrow a$

19. $S \rightarrow AbSA$

20. $S \rightarrow abC$

21. $D \rightarrow a$

22. $S \rightarrow bSA$

23. $S \rightarrow AbS$

- | | |
|-------------------------|--------------------------|
| 7. $B \rightarrow AbSA$ | 24. $S \rightarrow bS$ |
| 8. $C \rightarrow abC$ | 25. $D \rightarrow bSA$ |
| 9. $D \rightarrow AB$ | 26. $D \rightarrow AbS$ |
| 10. $S \rightarrow b$ | 27. $D \rightarrow bS$ |
| 11. $S \rightarrow a$ | 28. $D \rightarrow AbSA$ |
| 12. $A \rightarrow BAb$ | |
| 13. $A \rightarrow ABb$ | |
| 14. $A \rightarrow Bb$ | |
| 15. $B \rightarrow bSA$ | |
| 16. $B \rightarrow AbS$ | |
| 17. $B \rightarrow bS$ | } |

3. Elimination of nonproductive symbols.

$$PROD(G) = \{A \mid A \in V_N, \exists A \Rightarrow v, v \in V_T\}$$

$$NEPROD(G) = V_N \setminus PROD(G)$$

$$V_N = \{S, A, B, C, D\}$$

$$PROD(G) = \{B, S, A, D\}$$

$$NEPROD(G) = \{S, A, B, C, D\} \setminus \{B, S, A, D\} = \{C\}$$

- $P''' = \{$
- | | |
|-------------------------|--------------------------|
| 1. $S \rightarrow a$ | |
| 2. $S \rightarrow bA$ | 18. $S \rightarrow AbSA$ |
| 3. $S \rightarrow aA$ | |
| 4. $A \rightarrow aS$ | 19. $D \rightarrow a$ |
| 5. $A \rightarrow ABAb$ | 20. $S \rightarrow bSA$ |
| 6. $B \rightarrow a$ | 21. $S \rightarrow AbS$ |
| 7. $B \rightarrow AbSA$ | 22. $S \rightarrow bS$ |
| 8. $D \rightarrow bSA$ | |
| 9. $D \rightarrow AB$ | 23. $D \rightarrow AbS$ |
| 10. $S \rightarrow b$ | 24. $D \rightarrow bS$ |
| 11. $S \rightarrow a$ | 25. $D \rightarrow AbSA$ |
| 12. $A \rightarrow BAb$ | |
| 13. $A \rightarrow ABb$ | |
| 14. $A \rightarrow Bb$ | |
| 15. $B \rightarrow bSA$ | |
| 16. $B \rightarrow AbS$ | |
| 17. $B \rightarrow bS$ | } |

4. Elimination of inaccessible symbols:

$$\text{Initial } ACCES(G) = \{S\}$$

$$ACCES(G) = \{x \mid \exists S \Rightarrow \alpha x \beta\}$$

$$INACCES(G) = (V_N \cup V_T) \setminus ACCES(G)$$

$$ACCES(G) = \{S, A, b, a, B\}$$

$$V_N = \{S, A, B, D\} \quad V_T = \{a, b\}$$

$$INACCES(G) = \{S, A, B, D, a, b\} \setminus \{S, A, b, a, B\} = \{D\}$$

$P^{IV} = \{$
 1. $S \rightarrow a$
 2. $S \rightarrow bA$
 3. $S \rightarrow aA$
 4. $A \rightarrow aS$
 5. $A \rightarrow ABAb$
 6. $B \rightarrow a$
 7. $B \rightarrow AbSA$
 10. $S \rightarrow b$
 11. $S \rightarrow a$
 12. $A \rightarrow BAb$
 13. $A \rightarrow ABb$
 14. $A \rightarrow Bb$
 15. $B \rightarrow bSA$
 16. $B \rightarrow AbS$
 17. $B \rightarrow bS$
 18. $S \rightarrow AbSA$
 19. $S \rightarrow bSA$
 20. $S \rightarrow AbS$
 21. $S \rightarrow bS$
 $\}$

5. The Chomsky Normal Form

A grammar in the Chomsky Normal Form is a grammar of rules that has a form $A \rightarrow BC, D \rightarrow i$, where $A, B, C, D \in V_N$ and $i \in V_T$

$P^V = \{$
 1. $S \rightarrow X_1A$
 2. $S \rightarrow X_2A$
 3. $A \rightarrow X_2S$
 4. $A \rightarrow AY_1$
 5. $B \rightarrow a$
 6. $B \rightarrow A Y_3$
 7. $S \rightarrow b$
 8. $S \rightarrow a$
 9. $A \rightarrow BY_2$
 10. $A \rightarrow AY_5$
 11. $A \rightarrow BX_1$
 12. $B \rightarrow X_1Y_4$
 13. $B \rightarrow AY_6$
 14. $B \rightarrow X_1S$
 17. $S \rightarrow X_1Y_4$
 18. $S \rightarrow AY_6$
 19. $S \rightarrow X_1S$
 20. $X_1 \rightarrow b$
 21. $X_2 \rightarrow a$
 22. $Y_1 \rightarrow B Y_2$
 23. $Y_2 \rightarrow AX_1$
 24. $Y_3 \rightarrow X_1Y_4$
 25. $Y_4 \rightarrow SA$
 26. $Y_5 \rightarrow BX_1$
 27. $Y_6 \rightarrow X_1S$
 $\}$

$V_N = \{S, A, B, X_1, X_2, Y_1, Y_2, Y_3, Y_4, Y_5, Y_6\}$

$G=(V_N, V_T, P, S)$ $V_N=\{S, H, B\}$ $V_T=\{0, 1\}$

$P=\{$
1. $S \rightarrow 0S1$
2. $S \rightarrow 0SH$
3. $S \rightarrow 0$
4. $S \rightarrow 1B0$
5. $B \rightarrow 1H0$
6. $B \rightarrow SH$
7. $H \rightarrow 1AB$
8. $A \rightarrow SB$
 $\}$

1. Inccesible = $\{\}$
2. Productive= $\{S\}$
Non-productive= $\{A, B, H\}$

$P'=\{$
1. $S \rightarrow 0S1$
2. $S \rightarrow 0$
 $\}$

$P''=\{$
 $S \rightarrow X1X3$
 $X1 \rightarrow 0$
 $X2 \rightarrow 1$
 $X3 \rightarrow SX2$
 $S \rightarrow 0\}$

Chomsky Normal Form rules:

$N \rightarrow NN$

$N \rightarrow T$

Greibach Normal Form rules:

$N \rightarrow TN^*$

$N \rightarrow T$